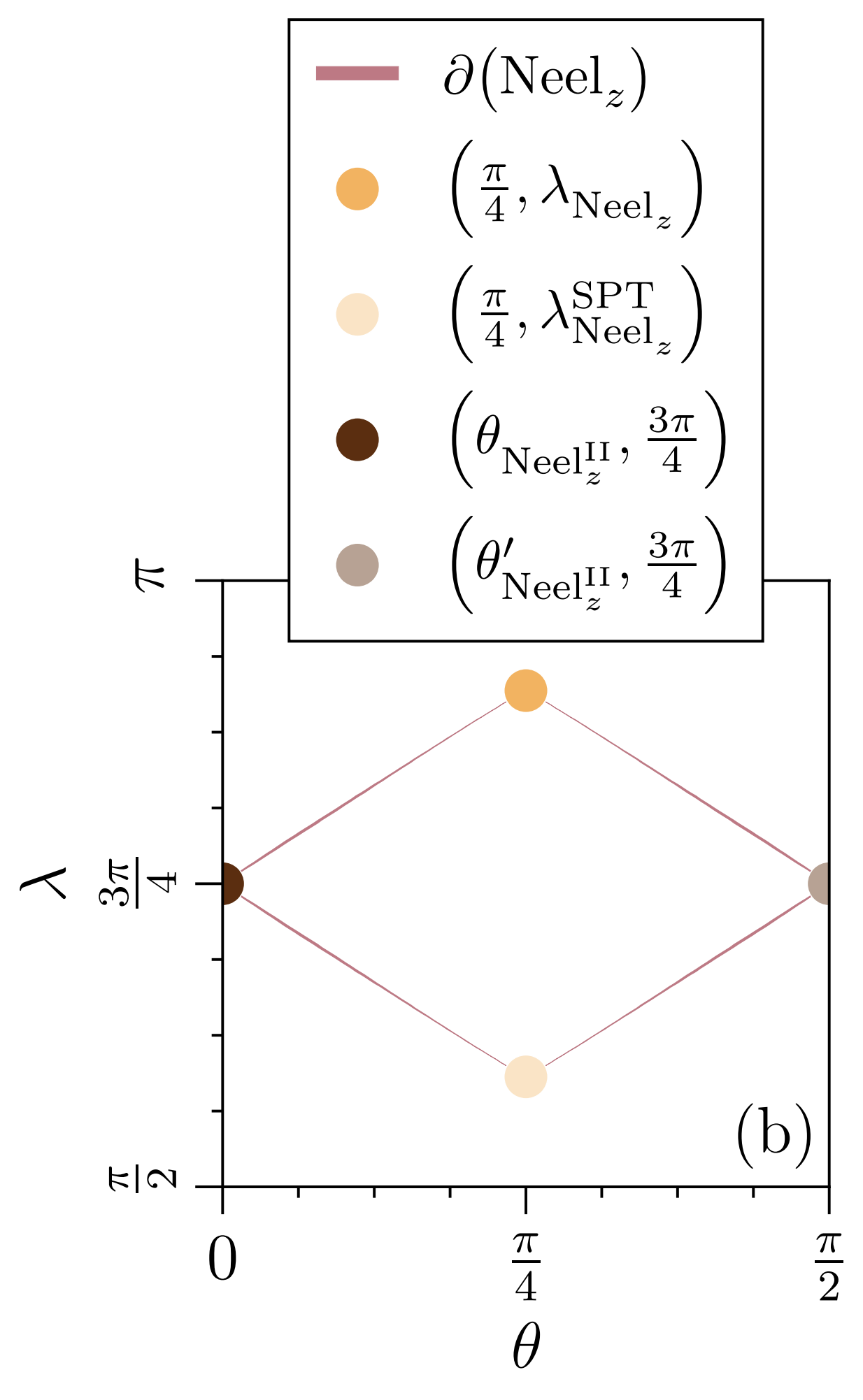

$\partial(\mathrm{Neel}_z)$
$\left(\frac{\pi}{4}, \lambda_{\mathrm{Neel}_z}\right)$
$\left(\frac{\pi}{4}, \lambda^{\mathrm{SPT}}_{\mathrm{Neel}_z}\right)$
$\left(\theta_{\mathrm{Neel}_z^{\mathrm{II}}}, \frac{3\pi}{4}\right)$
$\left(\theta'_{\mathrm{Neel}_z^{\mathrm{II}}}, \frac{3\pi}{4}\right)$
π
$\frac{3\pi}{4}$
$\frac{\pi}{2}$
λ
0
$\frac{\pi}{4}$
$\frac{\pi}{2}$
θ
(b)